

FMF — Mediterranean Pharmacy Stall (Pyrin, Attacks & Amyloidosis)





## 1. Mediterranean flags

**WHAT YOU SEE**

Armenian, Turkish, Israeli flags over the bazaar

**WHAT IT ENCODES**

FMF is the most common monogenic autoinflammatory (hereditary periodic fever) disease and clusters in eastern Mediterranean ancestries — Sephardic/North African and Iraqi Jews, Armenians, Turks, and Arabs — with carrier frequencies up to 1 in 5 in some groups. It is highly relevant in Israel and is a favorite board setup: a young patient of Mediterranean descent with recurrent self-limited fevers. Onset is usually in childhood/adolescence, and >90% present before age 20.

**BOARD TRAP**

A board stem describing recurrent fevers in a Northern European / Scandinavian child is steering you AWAY from FMF — pick TRAPS (which is autosomal dominant and common in those populations) or another periodic fever. Ancestry is a deliberate discriminator on the exam.

## 2. MEFV / pyrin pot

**WHAT YOU SEE**

Chef labeled PYRIN stirring a glowing cauldron

**WHAT IT ENCODES**

The MEFV gene (chromosome 16) encodes pyrin (marenostrin), expressed in neutrophils/monocytes. Loss-of-function pyrin mutations (M694V is the most common and most severe, associated with higher amyloidosis risk) remove pyrin's normal brake on the inflammasome, leading to unregulated assembly and uncontrolled IL-1β. Most pathogenic mutations sit in exon 10.

**BOARD TRAP**

The defect is in MEFV/pyrin, NOT in colchicine metabolism, NOT in NLRP3 (that is the cryopyrinopathies FCAS/MWS/NOMID), and NOT in mevalonate kinase (that is HIDS). Genetic testing confirms but is not required: FMF is a clinical diagnosis, and a negative MEFV panel does not exclude it.

## 3. Autosomal recessive

**WHAT YOU SEE**

AUTO-RECESSIVE banner with 1:10 carrier note

**WHAT IT ENCODES**

Classic FMF is autosomal recessive (two MEFV mutations), with very high carrier frequencies (~1:5 to 1:10) in at-risk groups so an affected sibling without affected parents is typical. A minority of single-mutation carriers can show milder/variable phenotypes, but boards expect recessive inheritance.

**BOARD TRAP**

Don't call it autosomal dominant — that is the trap meant to confuse it with TRAPS (TNFRSF1A, dominant). Likewise X-linked points to VEXAS, not FMF.

## 4. Caspase-1 / IL-1β

**WHAT YOU SEE**

CASP-1 and IL-1β labels by the flame

**WHAT IT ENCODES**

Mutant pyrin permits unchecked inflammasome activation → caspase-1 (the cysteine protease) cleaves pro-IL-1β into active IL-1β, driving neutrophilic serosal inflammation and the febrile attacks. This IL-1 dependence is exactly why IL-1 inhibitors (anakinra, canakinumab) work in colchicine-resistant disease.

**BOARD TRAP**

The dominant cytokine is IL-1β, NOT TNF-α. A stem suggesting a TNF inhibitor (etanercept/infliximab) as the rational mechanism is wrong for FMF — that targeting belongs to TRAPS/DADA2, not FMF.

## 5. Recurrent short fever

**WHAT YOU SEE**

Clock 24–72h and a STOP sign with self-limited man

**WHAT IT ENCODES**

Attacks are recurrent, stereotyped, self-limited febrile episodes lasting 1–3 days (12–72h) that resolve spontaneously, separated by symptom-free intervals of weeks to months. During attacks inflammatory markers spike (ESR, CRP, leukocytosis, SAA) and then normalize. Triggers include stress, exertion, menses, and cold.

**BOARD TRAP**

Fever persisting >7 days argues AGAINST FMF and toward TRAPS (attacks last >1 week, often weeks). HIDS attacks run 3–7 days. If you anchor on duration, you can sort the periodic fevers correctly — FMF is the shortest of the classic three.

## 6. Peritonitis (acute abdomen)

**WHAT YOU SEE**

Man clutching belly, FREE FLUID and 'mimics surgery' tags

**WHAT IT ENCODES**

Sterile peritonitis is the most common serositis (~90% of patients) — severe diffuse abdominal pain with guarding, rebound, and even free fluid that mimics a surgical acute abdomen, so patients classically undergo unnecessary appendectomies/laparotomies. Pain is self-limited and resolves within 1–3 days; recurrent attacks can leave peritoneal adhesions.

**BOARD TRAP**

The WRONG move is to take the patient to the OR for appendicitis. The discriminator: recurrent, stereotyped, self-resolving episodes with prior identical attacks and Mediterranean ancestry point to FMF serositis, not a surgical abdomen — diagnostic colchicine response supports it.

## 7. Pleuritis

**WHAT YOU SEE**

A man clutching one side of his chest with a 'PLEURITIS' tag — the serosal lining of the lung is inflamed, giving sharp one-sided pleuritic pain.

**WHAT IT ENCODES**

Pleuritis produces acute unilateral pleuritic chest pain from serosal inflammation, sometimes with a small exudative effusion and decreased breath sounds, resolving in 1–3 days. Pericarditis can occur but is much less common.

**BOARD TRAP**

Unilateral pleuritic chest pain in a young Mediterranean patient with recurrent self-limited attacks is FMF serositis — NOT pulmonary embolism (no Wells-criteria risk, no hypoxia/tachycardia pattern) and NOT pneumonia (no infiltrate/sputum, sterile and self-limited).



## 8. Erysipeloid rash

### WHAT YOU SEE

Sharp red erysipelas-like patch on the lower leg

### WHAT IT ENCODES

Erysipeloid erythema — a sharply demarcated, hot, tender red plaque over the ankle, dorsum of the foot, or shin — is the pathognomonic FMF skin sign during attacks and clears with the attack in days. It reflects the underlying serosal/neutrophilic inflammation rather than infection.

### BOARD TRAP

The WRONG answer is bacterial cellulitis/erysipelas treated with antibiotics. Discriminator: the FMF lesion is sterile, recurs in the same distribution, accompanies fever and serositis, and self-resolves in 1–3 days — antibiotics do nothing and cultures are negative.

## 9. Monoarthritis

### WHAT YOU SEE

Swollen single large joint figure

### WHAT IT ENCODES

Acute self-limited monoarthritis of a large weight-bearing joint (knee, ankle, hip) is the typical articular attack — markedly inflamed, sterile synovial fluid, resolving over days to a couple of weeks without joint destruction. A protracted (chronic) arthritis variant exists but is uncommon.

### BOARD TRAP

FMF arthritis is monoarticular, non-erosive, and self-limited — do NOT mistake it for rheumatoid arthritis (chronic symmetric small-joint polyarthritis with erosions/RF/anti-CCP) or septic arthritis (positive cultures, single episode that does not spontaneously remit and recur).

## 10. Elevated SAA

### WHAT YOU SEE

An 'SAA' label with an arrow feeding directly into the red 'AA AMYLOID' flag — serum amyloid A is the precursor protein that, when chronically high, misfolds into the amyloid fibrils.

### WHAT IT ENCODES

Serum amyloid A (SAA) is the acute-phase precursor protein driven up by chronic IL-1-mediated inflammation. Persistently elevated SAA — even subclinically between attacks — is what fuels misfolded AA fibril deposition; the goal of therapy is to keep SAA/CRP normalized, not merely to abort attacks.

### BOARD TRAP

SAA is the AA-amyloid precursor. The danger is sustained SAA elevation BETWEEN attacks (ongoing subclinical inflammation), not just the peak during an attack — which is why colchicine is taken continuously and dosed to normalize markers.

## 11. AA amyloidosis (renal)

### WHAT YOU SEE

Red AA AMYLOID flag, 'PROTEINURIA between attacks RED FLAG'

### WHAT IT ENCODES

AA (secondary) amyloidosis is the most feared complication: SAA-derived fibrils deposit in the kidney first, causing proteinuria → nephrotic syndrome → progressive renal failure/ESRD, and can also involve gut, liver, and spleen. Biopsy shows Congo-red-positive, apple-green birefringent deposits. M694V homozygosity and untreated disease raise the risk.

### BOARD TRAP

The feared complication is AA amyloidosis, NOT AL (light-chain) amyloidosis — AL is the plasma-cell/myeloma type and is the wrong choice here. New proteinuria, especially persisting between attacks, is the red-flag screen for amyloid renal involvement and an indication to confirm and escalate therapy.

## 12. Colchicine — daily lifelong

### WHAT YOU SEE

COLCHICINE pharmacy stall, DAILY (not PRN) sign

### WHAT IT ENCODES

Colchicine is first-line, taken DAILY and lifelong (typical adult dose ~1.2–1.8 mg/day, titrated; pediatric dosing weight-based). It inhibits microtubule polymerization and neutrophil chemotaxis, reducing both attack frequency AND amyloidosis risk. It is continued in pregnancy and lactation. GI upset is the main side effect; reduce dose in renal/hepatic impairment and beware interaction with CYP3A4/P-gp inhibitors (clarithromycin, statins, cyclosporine) which raises toxicity risk.

### BOARD TRAP

Colchicine is PROPHYLACTIC and continuous — it is NOT taken only during an attack (not PRN) and does not reliably abort an attack in progress. Stopping it because the patient feels well is the wrong move; lifelong adherence is what prevents amyloidosis.

## 13. Colchicine prevents amyloid

### WHAT YOU SEE

Green 'PREVENTS AMYLOID' arrow from the stall

### WHAT IT ENCODES

Beyond symptom control, colchicine's key disease-modifying benefit is preventing AA amyloidosis and preserving renal function by suppressing SAA. Adherent patients have dramatically lower rates of amyloid/ESRD, which is why it is given even to patients with few or mild attacks.

### BOARD TRAP

Even when attacks are infrequent or mild, lifelong colchicine is still indicated — to prevent amyloidosis. Withholding it in 'mild' disease is a board trap; the renal protection, not just symptom relief, drives the recommendation.

## 14. IL-1 inhibitors 2nd line

### WHAT YOU SEE

IL-1 INHIBITORS 2ND LINE shelf (canakinumab/anakinra)

### WHAT IT ENCODES

For colchicine-resistant (~5–10% of patients) or colchicine-intolerant FMF, IL-1 inhibitors are second-line: anakinra (daily subcutaneous IL-1 receptor antagonist) or canakinumab (monthly anti-IL-1 $\beta$  monoclonal). They reduce attacks and SAA in refractory disease; colchicine is generally continued alongside them.

### BOARD TRAP

IL-1 blockers are SECOND-line — colchicine stays first-line and remains the amyloid-preventer. Reaching for a biologic before an adequate colchicine trial (or stopping colchicine once a biologic is started) is the wrong sequence on the exam.